Rangan Pratik Borah

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LinkedIn GitHub

CAREER OBJECTIVE

Eager to contribute to real-world projects in electronics, embedded systems, web development or AI/software development while continuously learning and improving technical skills to gain practical industry exposure.

TECHNICAL SKILLS

Programming Languages (Python, Java) | **AI & Machine Learning** (OpenCV, NumPy, Haar Cascade, LBPH, 128-D feature vectors) | **Embedded Systems** (Arduino, ESP32, Raspberry Pi, sensor integration, IoT automation) | **Web Development** (HTML, React + Tailwind) | **Databases** (SQL (SQLite / MySQL) | **Hardware & Design** (Logic Design, Intel Quartus Prime, Electronics fundamentals) | **Tools & Platforms** (GitHub, VS Code, Arduino IDE, Eclipse IDE, Spyder) |

SOFT SKILLS

Problem-solving, Teamwork, Hackathon experience, Quick learner, Adaptable

EDUCATION

- **Bachelor of Technology - Electronics & Communication Engineering** | 2025 (currently completed Semester I)
  - *Vellore Institute of Technology (VIT), Vellore | GPA: 8.6 / 10 (Semester I)*
  - Relevant coursework: AI/ML, Python, Java, Logic Design, Management
- Class XII - South Point School, Guwahati | Percentage: 83% | 2025
- Class X - Ramanujan Senior Secondary School | Percentage: 95.33% | 2023

PROJECTS

1. MouFace NEXUS AI – Face Recognition System *(Duration: 4 Days)*

GitHub LINK **ngrok web LINK**

Tech Stack: Java, Spring Boot, OpenCV, MySQL, Haar Cascade, LBPH, HTML, CSS, JavaScript

Description:

- Built a real-time face recognition system with web-based registration and live detection
- Used Haar Cascade for detection and LBPH for recognition with optimized accuracy
- Integrated MySQL for user data management and real-time retrieval
- Developed full pipeline: capture - training - recognition - display
- Deployed live demo access using **ngrok** for remote usage

2. Moodora AI – Emotion-Based Music Recommendation System *(Duration: 2 Days)*

GitHub LINK **Vercel web LINK**

Tech Stack: React, JavaScript, MediaPipe, Spotify API, CSS, Vercel

Description:

- Built a real-time web app that detects facial emotions and recommends music accordingly
- Integrated MediaPipe for face analysis and Spotify API for playback and playlist mapping
- Implemented manual mood selection with playback controls (play, pause, next, previous)
- Designed a responsive React UI and deployed the application on Vercel

3. AI-Based Smart Irrigation System *(Duration: 2 Days)*

GitHub LINK

Tech Stack: Arduino, ESP32, Sensors

Description:

- Developed an IoT-based irrigation system using real-time soil and environmental data
- Integrated AI to take watering decisions, reducing unnecessary water usage
- Implemented sensor integration and control logic on ESP32 for end-to-end automation

4. Passcode-Based Electronic Security System *(Duration: 1 Day)*

GitHub LINK

Tech Stack: Arduino, ESP 32

Description:

- Built a passcode-protected security system with real-time input validation
- Implemented hardware-software integration with feedback via buzzer and LEDs

5. Half Adder Circuit | Digital Logic Design *(Duration: 1 Day)*

Tech Stack: Intel Quartus Prime

Description:

- Designed and simulated a half adder circuit demonstrating combinational logic fundamentals

KEY ACHIEVEMENTS

- Built **AI + Embedded** Systems projects in semester I
- Awarded the prestigious **Memento of Honour** at the UN Geneva International Exhibition
- Participated in Paintathon and received certificate from **World Book of Records**, London
- Developed a real-time face recognition system with 90%+ accuracy
- Participated in 2 hackathons on JAVA programming and IoT, and received certificate from VIT-TBI

INTERESTS & CO-CURRICULAR

- Robotics Club Member - Worked on robotics hardware, embedded systems and engineering projects
- Digital Art & Design - Skilled in digital art, graphic design and video editing
- Fine Arts Background - Completed Surnandan Bharati's 8-year art program; awarded the **Chittrasree Award** and **Sishuchitranandan Award** for excellence in painting
- International Exposure - Participated in multiple international and national art exhibitions